

FUKUSHIMA, Japan

WMO: 475950

Lat: 37.7584N

Lon: 140.4710E

Elev: 227

StdP: 14.58

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.0	26.9	13.9	11.2	37.5	15.9	12.4	36.0	18.6	36.9	16.7	37.1	4.0	270	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.2	93.6	76.8	90.9	75.9	88.0	74.4	78.3	89.4	77.3	87.5	76.3	85.3	7.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	134.6	83.2	74.4	129.8	82.4	73.5	125.5	81.7	42.1	90.4	41.0	87.8	39.9	85.5	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.6	14.6	13.1	DB	21.8	97.7	1.9	1.8	20.5	99.0	19.4	100.0	18.4	101.0	17.0	102.4	(4)
			WB	20.0	80.3	1.7	1.1	18.8	81.1	17.8	81.8	16.9	82.4	15.7	83.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	35.5	36.9	43.0	53.6	63.7	69.9	77.0	79.3	71.7	61.0	49.5	40.3	(6)
(7)		DBStd	16.13	4.02	4.74	5.86	6.70	5.79	5.22	5.95	5.69	6.05	5.46	5.72	4.87	(7)
(8)		HDD50	1470	448	366	232	39	1	0	0	0	1	79	304	(8)	
(9)		HDD65	4228	913	785	683	346	93	16	1	0	10	149	465	767	(9)
(10)		CDD50	3990	0	1	14	146	425	598	838	909	652	341	64	2	(10)
(11)		CDD65	1274	0	0	0	3	52	164	375	444	212	24	0	0	(11)
(12)		CDH74	10157	0	0	0	61	441	1006	3167	4079	1327	76	0	0	(12)
(13)		CDH80	3705	0	0	0	10	97	265	1249	1697	381	6	0	0	(13)
(14)	Wind	WSAvg	5.3	5.6	5.9	6.2	6.3	5.8	5.0	4.6	4.3	4.4	4.8	5.3	(14)	
(15)	Precipitation	PrecAvg	47.9	2.5	2.0	2.9	3.1	3.4	4.9	6.7	6.0	6.1	4.7	2.9	2.6	(15)
(16)		PrecMax	61.8	5.1	4.0	5.8	5.7	5.7	8.3	12.4	20.7	11.7	14.1	5.6	6.6	(16)
(17)		PrecMin	33.2	1.0	0.6	0.4	0.1	0.8	2.5	1.9	1.2	0.6	0.7	0.4	0.9	(17)
(18)		PrecStd	6.9	1.2	0.8	1.5	1.7	1.4	1.9	2.8	4.2	3.2	3.5	1.3	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	52.2	58.3	69.3	80.6	87.4	89.8	96.0	97.1	91.9	80.1	68.9	59.1	(19)	
(20)		MCWB	44.5	46.7	51.5	58.6	64.7	70.1	77.3	77.6	75.3	67.8	57.2	50.9	(20)	
(21)		DB	47.9	52.7	63.0	74.5	82.5	86.1	93.3	94.5	87.9	76.0	65.3	54.6	(21)	
(22)		MCWB	40.3	43.6	48.8	56.2	61.9	68.5	77.1	77.2	74.2	64.7	55.5	47.0	(22)	
(23)		DB	44.9	48.7	58.2	70.1	79.0	83.2	90.2	91.9	84.4	72.8	62.1	51.3	(23)	
(24)		MCWB	38.2	40.4	47.1	54.5	61.2	67.8	75.8	76.8	72.1	62.6	53.7	44.6	(24)	
(25)		DB	42.6	45.4	54.0	66.2	75.4	80.0	87.1	89.0	81.1	70.2	59.4	48.6	(25)	
(26)		MCWB	36.8	38.2	44.7	52.6	59.9	66.9	74.6	76.0	70.9	61.2	52.4	42.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	45.4	49.4	54.8	64.6	68.9	74.5	79.7	80.1	77.4	70.5	60.6	52.0	(27)	
(28)		MCDWB	50.8	57.4	63.8	72.6	79.6	83.1	92.0	91.6	86.8	77.1	65.5	56.7	(28)	
(29)		WB	41.3	44.3	51.3	59.3	66.1	72.4	78.1	78.9	75.7	67.1	57.7	48.2	(29)	
(30)		MCDWB	46.5	51.0	59.4	69.9	76.1	80.7	89.5	90.0	84.1	72.6	62.2	53.7	(30)	
(31)		WB	39.0	41.4	48.4	56.7	64.1	70.6	77.1	77.9	74.4	64.8	55.4	45.2	(31)	
(32)		MCDWB	43.7	47.7	56.8	65.4	73.4	78.6	87.4	88.3	81.9	70.2	60.8	50.7	(32)	
(33)		WB	37.1	38.8	45.1	54.5	62.3	69.0	75.9	76.9	73.0	62.8	53.0	42.8	(33)	
(34)		MCDWB	42.0	44.7	53.1	63.2	71.3	76.7	84.7	86.5	79.5	68.6	58.1	48.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	11.8	13.6	16.4	18.7	18.5	15.4	13.9	14.2	13.7	14.8	15.2	12.8	(35)	
(36)		5% DB	16.3	19.9	25.1	26.4	25.8	21.5	19.2	18.5	17.5	17.9	19.6	17.6	(36)	
(37)		MCWBR	10.6	12.5	14.5	12.6	10.5	8.5	6.9	6.1	7.0	9.0	11.6	11.6	(37)	
(38)		5% WB	15.5	19.0	22.6	21.0	19.9	16.5	17.1	16.8	15.2	13.8	17.3	17.1	(38)	
(39)		MCWBR	10.9	12.8	14.7	11.9	10.1	7.5	6.8	6.1	6.9	8.4	11.9	12.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.426	0.489	0.578	0.636	0.654	0.664	0.670	0.659	0.563	0.504	0.456	0.428	(40)	
(41)		taud	1.915	1.775	1.666	1.615	1.620	1.657	1.697	1.716	1.890	1.952	1.991	1.982	(41)	
(42)		Ebn at Noon	223	224	217	215	215	213	210	207	220	219	212	212	(42)	
(43)		Edh at Noon	45	59	73	81	82	79	75	72	57	49	42	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	657	891	1181	1452	1620	1526	1398	1408	1153	929	692	563	(44)	
(45)		RadStd	44	58	80	140	165	104	220	175	93	91	45	41	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	(46)
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page